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ChemComm

Cite this: *Chem. Commun.*, 2012, **48**, 12074–12076

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$FeCl_3$-catalyzed cyclization of α-sulfonamido-allenes with aldehydes—the substituent effect†

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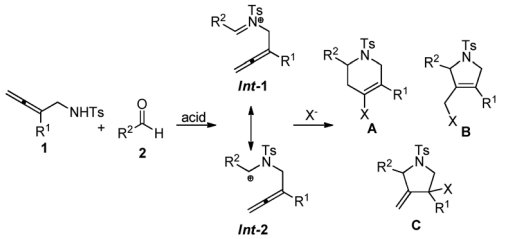
Received 25th September 2012, Accepted 19th October 2012
DOI: 10.1039/c2cc36941a

Published on 22 October 2012 on http://pubs.rsc.org | doi:10.1039/C2CC36941A
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$FeCl_3$-catalyzed aza-Prins-cyclization reaction of α-sulfonamido-allenes with aldehydes afforded 1,2,3,6-tetrahydropyridine or 2,5-dihydro-1*H*-pyrrole derivatives efficiently and highly selectively. The different regioselectivity is probably caused by the stabilizing effect of the phenyl group on the positively charged allylic intermediate.

Functionalized 1,2,3,6-tetrahydropyridines[1] are versatile building blocks widely utilized in the total synthesis of natural products[2] and this important structural motif widely exists in compounds displaying significant pharmacological and biological activities.[3] The cyclic double bonds are not only responsible for their interesting biological properties but also serve as a functional group for further manipulations in organic synthesis. Various methodologies have been developed toward the construction of 1,2,3,6-tetrahydropyridine derivatives, including Ru-catalyzed olefin metathesis,[4] palladium-,[5] silver-catalyzed[6] cyclization reactions and various multistep syntheses.[7] However, these types of reactions usually require expensive transition-metal catalysts or strict reaction conditions such as anhydrous and anaerobic conditions, which restrict their applications. Thus, the development for highly efficient synthesis of 1,2,3,6-tetrahydropyridine derivatives under mild conditions are still in great demand.

Prins cyclization utilizing alkenes and alkynes as substrates has now emerged as a powerful tool for the synthesis of heterocycles.[8] On the other hand, the cyclization reaction of allenes has been extensively developed as an efficient methodology for the synthesis of cyclic products.[9] Based on these facts, we envisioned that the reaction of α-sulfonamido-allenes with aldehydes[10] under acid catalysis would provide imine intermediates ***Int*-1** or ***Int*-2** (Scheme 1). Sequential cyclization and nucleophilic attack may, in principle, provide three types of products **A–C**. Herein, we report a mild and highly selective aza-Prins-cyclization for efficient assembly of 1,2,3,6-tetrahydropyridine or 2,5-dihydro-1*H*-pyrrole skeletons from α-sulfonamido-allenes and aldehydes, depending on the nature of the R^1 substituent on the allene unit.



Scheme 1 Possible reaction pathways.

Our initial work began with *N*-(buta-2,3-dienyl)-4-tolylsulfonamide **1a** and 1-naphthaldehyde **2a** under the catalysis of Fe(III) and TMSCl.[11] As a first attempt, we were happy to notice that the reaction of **1a** (1 equiv.), **2a** (1.5 equiv.), $FeCl_3$ (20 mol%) and TMSCl (1.5 equiv.) in CH_2Cl_2 with stirring at 30 °C for 16 h afforded the cyclized product **3a** in 38% yield (Table 1, entry 1). Increasing the amount of catalyst improved the yield greatly (Table 1, entries 2 and 3). The yield was improved to 71% when reducing the loading of the aldehyde to 1.2 equiv. However, when we reduced the amount of aldehyde further, the yield dropped (Table 1, entries 4 and 5). Addition of anhydrous $MgSO_4$ led to no improvement (Table 1, entry 6). Then, the reaction was tested with different Fe(III) salts which showed that other Fe(III) salts were not as efficient as $FeCl_3$ (Table 1, entries 7 and 8). Efforts trying to reduce the loading of TMSCl were also not successful (Table 1, entries 9 and 10). The reaction did not take place if $FeCl_3$ was not added or using CH_3CN as the solvent (Table 1, entries 11–13). Thus, $FeCl_3$ (45 mol%), aldehyde (1.2 equiv.) and TMSCl (1.5 equiv.) in CH_2Cl_2 at 30 °C were defined as the optimized reaction conditions for further study.

With the optimized conditions in hand, we turned to examine the substrate scope of the reaction. Benzaldehyde (Table 2, entry 1) and aromatic aldehydes substituted by *o*-Cl, *p*-NO_2, *o*-Br groups on the phenyl ring all afforded good yields of the products (Table 2, entries 2–4). The reaction may also be extended to a variety of aliphatic aldehydes. In addition to primary- (Table 2, entries 5 and 6) and secondary-alkyl aldehydes (Table 2, entry 7), pivaldehyde

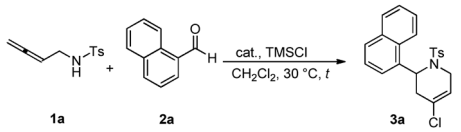
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† Electronic supplementary information (ESI) available: Experimental procedures, characterization data, and copies of ^1H and ^{13}C NMR spectra for all products. CCDC 880541 and 880542. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c2cc36941a

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Published on 22 October 2012 on http://pubs.rsc.org | doi:10.1039/C2CC36941A
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Table 1 Optimization of the reaction conditions for Fe(III)-catalyzed aza-Prins-cyclization of **1a** with **2a**[a]



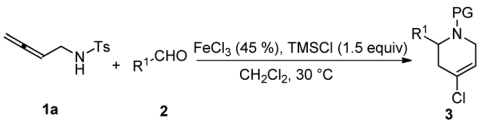
1a **2a** **3a**

Entry	Cat. (mol%)	**2a**/TMCl (equiv.)	*t*/h	Yield of **3a**[b] (%)	Recovery of **1a**[b] (%)
1	$FeCl_3$ (20)	1.5/1.5	21	38	19
2	$FeCl_3$ (30)	1.5/1.5	17	50	13
3	$FeCl_3$ (45)	1.5/1.5	21	60	10
4	$FeCl_3$ (45)	1.2/1.5	21	78 (71[c])	8
5	$FeCl_3$ (45)	1.0/1.5	21	67	9
6[d]	$FeCl_3$ (45)	1.2/1.5	22	73	0
7	$Fe(acac)_3$[e]	1.2/1.5	22	8	62
8	Fe(III)[f]	1.2/1.5	18	7	89
9	$FeCl_3$ (45)	1.2/—	21	48	35
10	$FeCl_3$ (45)	1.2/0.5	21	57	24
11	—	1.2/1.5	12	—	95
12[g]	—	1.2/1.5	18.5	—	98
13[g]	$FeCl_3$ (45)	1.2/—	12.5	—	96

[a] The reaction was conducted using **1a** (0.1 M), **2a** (1.2 equiv.), Fe(III) and TMSCl at 30 °C. [b] Determined by ^{1}H NMR analysis with mesitylene as the internal standard. [c] Isolated yield. [d] 1.0 equiv. $MgSO_4$ was used in this reaction. [e] 45 mol% of catalyst was used. [f] 45 mol% of $Fe(NO_3)_3{\cdot}9H_2O$ was used. [g] CH_3CN was used as solvent.

(Table 2, entry 8) and paraformaldehyde (Table 2, entry 9) may all used to afford the corresponding 1,2,3,6-tetrahydropyridine derivatives with excellent yields. The structure of **3j** was further confirmed by X-ray diffraction (see ESI†).[12]

Table 2 $FeCl_3$-catalyzed aza-Prins-cyclization of **1** with different aldehydes **2** under standard conditions[a]

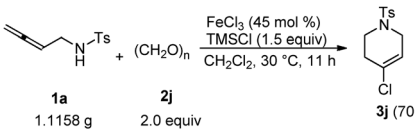


1a **2** **3**

Entry	R^1CHO	*t*/h	Yield of **3**[b] (%)
1	C_6H_5 (**2b**)	13	78 (**3b**)
2	2-ClC_6H_4 (**2c**)	13	80 (**3c**)
3	4-$NO_2C_6H_4$ (**2d**)	11	79 (**3d**)
4	2-BrC_6H_4 (**2e**)	11	83 (**3e**)
5	*n*-C_7H_{15} (**2f**)	7	83 (**3f**)
6	3-PhC_3H_6 (**2g**)	10	86 (**3g**)
7[c]	*c*-C_6H_{11} (**2h**)	8	84 (**3h**)
8[d]	*t*-Bu (**2i**)	12.5	58 (**3i**)[e]
9	H (**2j**)	11	74 (**3j**)

[a] The reaction was conducted at 30 °C in CH_2Cl_2 using **1a** (c = 0.1 M), aldehyde (1.2 equiv.), $FeCl_3$ (45 mol%) and TMSCl (1.5 equiv.). [b] Yield of isolated product. [c] 1.5 equiv. aldehyde was used. [d] 2.0 equiv. aldehyde was used. [e] The conversion of **1a** was 89% based on ^{1}H NMR analysis of the crude product with mesitylene as internal standard.

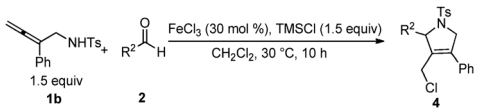
The reaction of **1a** and **2j** to afford **3j** in 70% yield can be readily carried out on a one gram scale (eqn (1)).



1a 1.1158 g **2j** 2.0 equiv **3j** (70 %) (1)

Then, *N*-(2-phenylbuta-2,3-dienyl)-4-tolylsulfonamide **1b** was applied as substrate to investigate a substituent effect. However, the Prins-type cyclization afforded the five-membered 2,5-dihydro-1*H*-pyrrole derivative **4a** exclusively. The structure was unambiguously established by X-ray diffraction (see ESI†).[13] As is well known, functionalized 2,5-dihydropyrrole is a commonly observed structural unit in various biologically active natural compounds.[14] Thus, we demonstrated that the $FeCl_3$-catalyzed aza-Prins-cyclization reaction of *N*-(2-phenylbuta-2,3-dienyl)-4-tolylsulfonamide **1b** and *p*-Br/F-benzaldehydes all proceeded smoothly to afford **4b** and **4c** in 66 and 74% yields, respectively (Table 3, entries 2 and 3). Aromatic aldehydes with electron-donating groups such as *p*-Me, *p*-MeO on the phenyl ring gave the products in excellent yields (Table 3, entries 4 and 5). Furthermore, *m*-MeO and *o*-F substituted aromatic aldehydes may also be used as substrates, although the yields are somewhat lower (Table 3, entries 6–7). The reaction may also be extended to aliphatic aldehydes such as **2q** which affords **4h** in 46% yield.

Table 3 $FeCl_3$-catalyzed aza-Prins-cyclization of **1b** with aldehydes **2** under standard conditions[a]

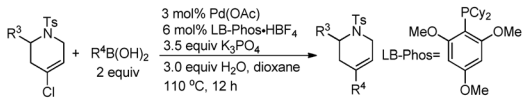


1b **2** **4**

Entry	R^2	Yield of **4**[b] (%)
1	4-ClC_6H_4 (**2c**)	67 (**4a**)
2	4-BrC_6H_4 (**2k**)	66 (**4b**)
3	4-FC_6H_4 (**2l**)	74 (**4c**)
4	4-MeC_6H_4 (**2m**)	81 (**4d**)
5	4-$MeOC_6H_4$ (**2n**)	67 (**4e**)
6[c]	3-$MeOC_6H_4$ (**2o**)	66 (**4f**)
7	2-FC_6H_4 (**2p**)	53 (**4g**)
8[d]	*n*-C_3H_7 (**2q**)	46 (**4h**)

[a] The reaction was carried out at 30 °C in CH_2Cl_2 using aldehyde (c = 0.1 M), **1b** (1.5 equiv.), $FeCl_3$ (30 mol%) and TMSCl (1.5 equiv.). [b] Yield of isolated product. [c] The reaction was stirred for 14 h. [d] The reaction was stirred for 36 h.

The 4-phenyl-1,2,3,6-tetrahydropyridine skeleton is a key structural motif in a series of bioactive molecules, such as Parkinsonian inducing neurotoxin 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP).[15] In order to show the potential of our method, the Suzuki coupling reactions of 1,2,3,6-tetrahydropyridine derivatives **3** with boronic acid were realized using the LB-Phos recently developed in this group as the ligand (Scheme 2).[16] The coupling of 1,2,3,6-tetrahydropyridine derivative **3j** with phenyl- and 2-furyl-boronic acid both afforded the cross-coupling products in excellent yields. This coupling is also suitable for **3b** to form the corresponding Suzuki coupling product **6c** in 85% yield.



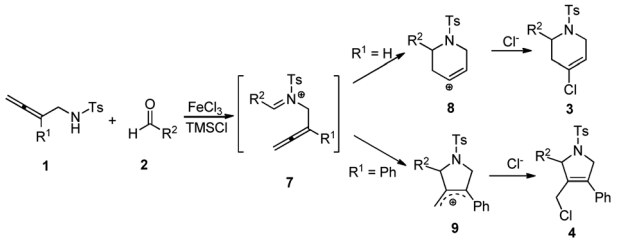
3 **5** **6**

R^3 = H (**3j**), R^4 = Ph (**5a**), **6a**, 85% yield;
R^3 = H (**3j**), R^4 = 2-furan (**5b**), **6b**, 90% yield;
R^3 = Ph (**3b**), R^4 = Ph (**5a**), **6c**, 85% yield.

Scheme 2 $Pd(OAc)_2$/LB-Phos-catalyzed Suzuki coupling of **3**.

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The high regioselectivity of this reaction may be explained as follows: the combination of $FeCl_3$ and TMSCl would work together to promote the condensation of the α-sulfonamido-allenes with aldehydes to afford the intermediate **7**. When R^1 = H, the regioselective cyclization of **7** occurs to afford the intermediate **8** which would finally lead to the formation of **3**. When R^1 = Ph, the positive charge of the intermediate **9** is stabilized and five-membered cyclic products **4** are formed exclusively (Scheme 3).



Scheme 3 Rationale for the selectivity observed.

In conclusion, we have demonstrated $FeCl_3$-catalyzed highly regioselective aza-Prins-cyclization reactions of α-sulfonamido-allenes with aldehydes. This reaction may produce 1,2,3,6-tetrahydropyridine or 2,5-dihydro-1*H*-pyrrole derivatives efficiently and highly selectively depending on the R^1 substituent on the allene unit. In view of the ready availability of the starting materials from terminal alkynes[17] and the mediators, this type of transformations may be useful in organic synthesis and medicinal chemistry. Further studies including mechanistic studies and the asymmetric variants of the cyclization reactions are being conducted in our laboratory.

Financial support from National Basic Research Program of China (Grant Nos. 21232006 and 2011CB808700) and National Natural Science Foundation of China (21172192) is greatly appreciated. We thank Mr Shangze Wu in our group for reproducing the results presented in entries 3, 6 of Table 2 and entry 3 in Table 3.

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12 Crystal data for **3j**. $C_{12}H_{14}ClNO_2S$, M_r = 271.75, monoclinic, space group $P2_1/c$, final R indices [$I > 2(I)$], R_1 = 0.0401, wR_2 = 0.1023, R indices (all data) R_1 = 0.0439, wR_2 = 0.1062, a = 7.7348(5), b = 8.2950(5), c = 20.1342(13) Å, β = 98.598(1)°, V = 1277.3(14) Å^3, T = 293 K, Z = 4, reflections collected/unique: 7493/2511 (R_{int} = 0.0285), number of observations [$I > 2(I)$], parameters: 156. CCDC 880541†.

13 Crystal data for **4a**. $C_{24}H_{21}Cl_2NO_2S$, M_r = 458.38, monoclinic, space group $P2_1/n$, final R indices [$I > 2(I)$], R_1 = 0.0536, wR_2 = 0.1300, R indices (all data) R_1 = 0.0759, wR_2 = 0.1405, a = 17.918(2), b = 5.8464(8), c = 21.295(3) Å, β = 99.577(4)°, V = 2199.7(5) Å^3, T = 296 K, Z = 4, reflections collected/unique: 24225/3857 (R_{int} = 0.0603), number of observations [$I > 2(I)$], parameters: 271. CCDC 880542†.

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